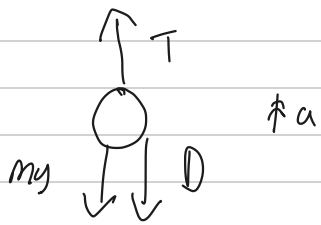
Equations:



Therefore

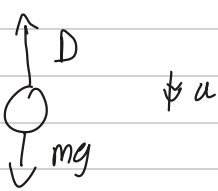
$$T - (mg + D) = ma$$

so

$$T - (mg + D) = m\frac{dv}{dt}$$ (LHS all forces) (RHS: $[M] \cdot [L][T]^{-2}$ which is dimension of force)

during the ascent

during descent:



maintain +ve as ↑

so

$$mg - D = m\frac{dv}{dt} \qquad -mg + D = m\frac{dv}{dt}$$

Initial condition:

at $t = 0$ and $v = 0$

$$T > mg$$

$$\Rightarrow \frac{T}{mg} > 1$$

this is known as the thrust to weight ratio and in practice is usually 5:1

in the first instant $T < mg$ so there will be a negative acceleration as thrust ramps up
so if $h=0$ and $a<0$, $a=0$

The definition for drag is:

$$D = \frac{1}{2} \rho C_d A v^2$$

where ρ = density of air
C_d = drag coefficient
A = reference area

Drag is $mv = -kv^2$ as during descent it doesn't change sign which makes drag act downwards

To maintain sign:

$D = \frac{1}{2} \rho C_d A v|v|$ where $|v|$ is speed + is upwards
- is downwards

$$\Rightarrow T - (mg + \frac{1}{2} \rho C_d A v|v|) = m \frac{dv}{dt}$$

where $k = \frac{1}{2} \rho C_d A$

Variable Mass

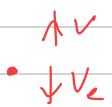
the rocket will lose mass as fuel as it flies

at time t : mass $= m$ velocity $= v$
at time dt : loses a mass dm
exhaust leaves at v_e relative to rocket (downwards)

Using cons of momentum:

at time t:

$p = mv$



So $v_{exhaust} = v_{rocket} + v_{exhaust\ rocket}$

$\Rightarrow v_{exhaust} = v - v_e$

at time $t+dt$:

two momentums: rocket and fuel

rocket loses mass

gain in velocity

momentum of the exhaust

So

$$p_{t+dt} = (m-dm)(v+dv) + dm(v-v_e)$$

dv appears from fuel being expelled

$$dp = p_{t+dt} - p_t$$

solving for dp:

$$dp = (m - dm)(v + dv) + dm(v - v_e) - mv$$
$$= mv + m\,dv - v\,dm - dm\,dv + v\,dm - v_e\,dm - mv$$
$$= m\,dv - dm\,dv - v_e\,dm$$

$dm\,dv$ is negligible (small terms multiplied)

$$= \underline{m\,dv - v_e\,dm} = dp$$

Using

$$p = \int f\,dt$$

we can write change in $p = f_{external} \cdot dt$

$$\Rightarrow \frac{dp}{dt} = f_{external}$$

$$\frac{1}{dt}(m\,dv - v_e\,dm) = -mg - kv|v|$$

$$m\frac{dv}{dt} - v_e\frac{dm}{dt} = -mg - kv|v|$$

$$\boxed{m\frac{dv}{dt} = v_e\frac{dm}{dt} - (mg + kv|v|)}$$

comparing to $m\frac{dv}{dt} = T - (mg + kv|v|)$

$$T = v_e\frac{dm_e}{dt} \quad \text{or} \quad T = v_e\dot{m}_e$$

Checking via units:

$$[L][T]^{-1} \cdot [M][T]^{-1}$$
$$= [M][L][T]^{-2} \text{ which are units of force!!!}$$

Checking against Tsiolkovsky:

assume in a vacuum ($g=0, h=0$)

$$\Rightarrow m\frac{dv}{dt} = v_e \dot{m}_e$$

$$m\frac{dv}{dt} = v_e \frac{dm_e}{dt}$$

Using $dm = -dm_e$

$$m\frac{dv}{dt} = -v_e \frac{dm}{dt}$$

$$m\,dv = -v_e\,dm$$

$$-\frac{1}{v_e}\,dv = \frac{1}{m}\,dm$$

$$-\frac{1}{v_e}v = \ln m + c$$

$$-v = v_e(\ln m + c)$$

at $v = v_0$ $m = m_0$

where v_0 is starting vel
m_0 is initial mass

$$-v_0 = v_e \ln m_0 + c$$

$$c = -v_0 - v_e \ln m_0$$

$$-v = v_e \ln m - v_0 - v_e \ln m_0$$

$$-v = v_e(\ln m - \ln m_0) - v_0$$

$$v_0 - v = v_e\left(\ln\left(\frac{m}{m_0}\right)\right)$$ (this is not Δv)

$$\Rightarrow v - v_0 = v_e \ln \frac{m_0}{m}$$

so $\boxed{\Delta v = v_e \ln \frac{m_0}{m}}$

A sim needs the mass at every time the data sheet gives:

- Thrust
- Impulse
- Propellant mass

Need to find $m(t)$

$$T = V_e \dot{m}_e$$

$$\frac{1}{V_e} T = \frac{dm_e}{dt}$$

$$\frac{1}{V_e} T\,dt = dm_e$$

→ assume V_e is constant

$$\frac{1}{V_e} \int_0^t T\,dt' = m_e$$

so

$$\frac{1}{V_e} \int_0^{t_b} T\,dt' = m_{ejected}(t)$$

at $t = t_{burnout}$, $m_{ejected}(t) = m_{propellant}$

→ also the total impulse

$$\Rightarrow \frac{1}{V_e} \int_0^{t_b} T\,dt' = m_{propellant}$$

$$\frac{1}{V_e} I_{total} = m_{propellant}$$

$$\frac{I_{total}}{m_{propellant}} = V_e$$

$$m(t) = m_0 - m_{ejected}(t)$$

at $t=0$ $m_{ejected}(0) = 0$ so $m(t) = m_0$

at $t = t_b$ $m_{ejected}(t) = m_{propellant}$ so $m(t) = m_0 - m_{prop}$

$$m(t) = m_0 - \frac{1}{V_e} I(t)$$

$$m(t) = m_0 - \frac{m_{propellant}}{I_{total}} I(t)$$

$$m(t) = m_0 - m_{propellant} \frac{I(t)}{I_{total}}$$

$$m'(t) = -\frac{m_{propellant}}{I_{total}} T(t)$$

Since $\Delta I = F \Delta t$

$$\Rightarrow \frac{\Delta I}{\Delta t} = F$$

Final results for the model:

$$\frac{dh}{dt} = v$$

$$\frac{dv}{dt} = \frac{T(t) - m(t)g - k\,v|v|}{m(t)}$$

$$m(t) = M_0 - m_{propellant} \frac{I(t)}{I_{total}}$$

Coast Phase

Is the part of flight with an exact solution used for a comparison

at burnout $t = t_b$

$t = 0$
$m = m_{dry}$
height $= h$
$v_1 > 0$

want to find coast height

$$\frac{dv}{dt} = \frac{dh}{dt}\cdot\frac{dv}{dh} = v\frac{dv}{dh}$$

$v > 0$ on way up so $v|v| = v^2$

$$m\frac{dv}{dt} = -(mg + kv^2)$$

$$mv\frac{dv}{dh} = -(mg + kv^2)$$

without drag

$$mv\frac{dv}{dh} = -mg$$

$$-\frac{v}{g}dv = dh$$

$$-\frac{v^2}{2g} = h + C$$

at $h = 0$ $v = v_b$

$$C = -\frac{v_b^2}{2g}$$

$$\frac{v_b^2 - v^2}{2g} = h$$

at apex $v = 0$

$$\Rightarrow \frac{v_b^2}{2g} = h$$

with drag:

$$m v \frac{dv}{dh} = -(mg + kv^2)$$

$$\frac{mv}{-(mg+kv^2)}\, dv = dh$$

$$\frac{mv}{-mg-kv^2}\, dv = dh$$

$$-\frac{1}{2}\frac{m}{k} \ln(mg+kv^2) = h + C$$

$$= -\frac{m}{2k} \ln(mg+kv^2) = h + C$$

at $h = 0$ $v = v_b$

$$-\frac{m}{2k} \ln(mg + k v_b^2) = C$$

$$\frac{m}{2k} \ln(mg + k v_1^2) - \frac{m}{2k} \ln(mg + kv^2) = h$$

$$\frac{m}{2k} \ln \frac{mg + k v_b^2}{mg + kv^2} = h \quad \text{at apex } v = 0$$

$$= \frac{m}{2k} \ln\left(\frac{mg + k v_b^2}{mg}\right) = h$$

$$\Rightarrow \frac{m}{2k} \ln\left(1 + \frac{k v_b^2}{mg}\right) = h$$

as $k \to 0$

$$\frac{m}{2k} \ln\left(1 + \frac{kv_1^2}{mg}\right) = h$$

$\frac{m}{2k} \to \infty$, $\frac{kv_1^2}{mg} \to 0$

$\frac{kv_b^2}{mg}$ becomes tiny call it x

$\ln(1+x)$

using maclaurin $\approx x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$ $(-1 < x \leq 1)$

if x is tiny $\frac{x^2}{2}$ is smaller so negligible

$\Rightarrow \ln(1+x) \approx x$

$\Rightarrow$ as $k \to 0$

$$\frac{m}{2k} \cdot \frac{kv_1^2}{mg} = \Delta h$$

$$= \frac{v_b^2}{2g} = \Delta h$$ Verified.

Choosing a Motor and Rocket to test

Motor: C6-7 motor

Rocket: Estes Alpha III

Specs:

$I_{total} = 8.8\ Ns$
$M_{prop} = 11\ g$
$M = 29\ g$
$t_b = 1.9\ s$
$T_{avg} = 4.7\ N$
$T_{peak} = 14.1\ N$
$D = 18\ mm$

So

$m_{dry} = m_{rocket} + 13$ $(29 - 11 = 13\ g$ of casing$)$
$m_0 = m_{rocket} + 29$

Rocket dimensions:

diameter $= r = 0.0125\ m$ (radii)
Mass (no motor) $= 0.039\ kg$
Max altitude $= 351\ m$
Parachute diameter $= 0.305\ m$

$A = \pi(0.0125)^2 = 4.91 \times 10^{-4}\ m^2$

(simulations done on open rocket)

Data to reference against

Apogee = 385.5
v_b = 112 ms^{-1}
h_b = 130 m
Time to apogee = 8.12s
v_{launch} = 20.2 ms^{-1}

Using Eulers Method

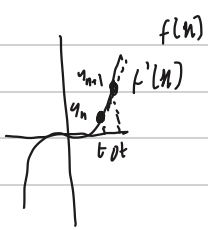
Used to approximate differential equations
e.g.

$$\frac{dy}{dt} = f(t,y), \quad y(0) = y_0$$

this is the rocket during the burn
the idea is at time step Δt assume gradient is constant
so:

$$y_{n+1} = y_n + \Delta t \cdot f(t_n, y_n)$$

or next value = old value + gradient × time step and iterate
graphically:



comes from the taylor series:

$$y(t+\Delta t) = y(t) + \Delta t\, y'(t) + \frac{1}{2}\Delta t^2 y''(t)$$

euler is a first order method
it drops the terms from Δt^2 onwards
thus

$$\text{error} = \Delta t^2$$

$$\text{number of steps} = \frac{T_{total}}{\Delta t}$$

so

$$O(\Delta t) = \frac{T_{total}}{\Delta t} \cdot \Delta t^2$$

and such:

$\frac{1}{2}\Delta t$ the error halves

the link:

3 quantities change with respect to time:

$$\frac{dh}{dt} = v$$

$$\frac{dv}{dt} = \frac{T - mg - kv|v|}{m}$$

$$\frac{dm}{dt} = -\frac{m_{prop}}{F_{total}} T$$

Using euler:

$$h_{n+1} = h_n + (v_n)\Delta t$$ → derivative of displacement

$$V_{n+1} = v_n + (a_n)\Delta t$$ → derivative of velocity

$$m_{n+1} = m_n + \left(\frac{dm}{dt}\right)_n \Delta t$$

Comparing euler against RK45

RK45 - Uses 4th and 5th orders to adaptively step
euler is first order so error $\propto \Delta t$
RK45 has a smaller error
makes use of the same state vector